



Syllabus

Spring Semester 2027 (샘플)

Course Title	Machine Learning and Deep Learning	Course Code	MC0000	Credit	3	Classroom	5701
		Type	Lecture	Hours	3		

Teacher	Name	Office	Office hour	Contact	E-mail
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1. Course Descriptions

- To understand and implement classical supervised learning (regression, classification) and unsupervised learning (clustering, dimensionality reduction) methods
- To correctly apply model selection and validation principles (regularization, cross-validation) to real data
- To understand the training principle behind neural networks (backpropagation) and design/implement representative deep learning architectures such as CNNs, RNNs, and Transformers
- To understand the basic principles of generative models
- To carry out the full pipeline of a machine learning project — problem definition, implementation, honest validation, and presentation — as a team

2. Core Target Competencies

Subject Core Competencies			
Fundamental Machine Learning and Problem-Solving Skills	Logical Thinking and Analytical Skills	Communication and Collaboration Skills	Ethical Use of AI and Responsible Attitude
• Demonstrates understanding of core machine learning and deep learning algorithms and applies them to analyze and solve real-world problems. • Derives the mathematical principles behind models (e.g. gradients, loss functions) and uses them to explain why a result occurred. • Clearly explains a team project's design and results to others, and gives constructive peer-review feedback to other teams. • Reports experimental results honestly (including unsuccessful attempts), and uses AI tools responsibly in coursework.			

3. Textbooks and References

A. Textbook: None (slides and handouts)

B. References

- "Machine Learning in Action", Peter Harrington

- "Machine Learning Design Patterns", V. Lakshmanan, S. Robinson, M. Munn
- "Deep Learning", Ian Goodfellow, Yoshua Bengio, Aaron Courville

★ There is no main textbook. Main materials can be downloaded from the Google Classroom.

4. Class Method

A. Google Classroom code : [TBD]

B. Lecture slides and code examples are distributed through the Google Classroom.

5. Evaluation

A. Evaluation Criteria

Quizzes & Homework	Midterm Exam.	Final Exam.	Team Project	Attendance and Attitudes	Sum
20	25	25	25	5	100

B. Evaluation Method

1. Absolute evaluation
2. Weekly quiz and homework for each class
3. Midterm exam: written test (2 hrs)
4. Final exam: written test (2 hrs)
5. Team project: two capstone projects (Week 8: classical ML, Week 16: deep learning), each graded on a written report, a presentation, and peer-review. Grading emphasizes the honesty of the experimental procedure (baseline comparison, proper train/validation/test separation, honest reporting of results) over raw performance.

6. Criteria for Awarding Credits

- **Pass:** In the case of receiving a score of **70 or more out of 100** according to the criteria of the evaluation method.
- **Academic Excellence Award:** Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.
- **Attendance:** At least **three-quarters** of the total class hours is required to earn credit.
- **Pass/Fail courses:** If the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- **Excused absences:** For official duties or medical reasons, the make-up score for a missed exam shall be calculated as follows: the percentile score corresponding to the student's total score in other assessments of the same course shall be applied to the missed exam, multiplied by a weighting factor (**100%** for official absences, **80%** for medical absences). If there are no available reference scores, a re-assessment shall be conducted.

※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

(Exact dates will be announced once the 2027 academic calendar is confirmed.)

Weeks	Date	Topic and Contents	Note
Week 1	TBD	Introduction to Machine Learning	
Week 2	TBD	Regression: Linear & Logistic Regression	
Week 3	TBD	Generative Classifiers: Naive Bayes, GDA	
Week 4	TBD	Distance-based Models: kNN, k-means, GMM/EM	
Week 5	TBD	Support Vector Machines and Kernels	
Week 6	TBD	Regularization and Model Selection	
Week 7	TBD	Tree-based Models: Decision Trees to GBDT	
Week 8	TBD	Block A Capstone: Team Project	Midterm exam
Week 9	TBD	Neural Network Basics, Backpropagation	
Week 10	TBD	Convolutional Neural Networks	
Week 11	TBD	Sequence Models (RNNs)	
Week 12	TBD	Attention and Transformers	
Week 13	TBD	Graph Representation Learning	
Week 14	TBD	Representation Learning: PCA, word2vec	
Week 15	TBD	Latent-Variable Generative Models	
Week 16	TBD	Block B Capstone: Team Project	Final exam

※ The above plans may change according to the academic schedule.